

A THEORY OF MORPHOLOGICAL NEURAL NETWORKS

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ABSTRACT

The theory of classical artificial neural networks has been used to solve pattern recognition problems in image processing that is different from traditional pattern recognition approaches. In standard neural network theory, the first step in performing a neural network calculation involves the linear operation of multiplying neural values by their synaptic strengths and adding the results. Thresholding usually follows the linear operation in order to provide for non-linearity of the network. This paper presents the fundamental theory for a morphological neural network which, instead of multiplication and summation, uses the non-linear operation of addition and maximum. Several basic applications which are distinctly different from pattern recognition techniques are given, including a net which performs a sieving algorithm.

1. Introduction. The recent resurgence of interest in artificial neural networks has brought a deluge of publications to the field [4,9,15]. Applications of neural networks are appearing in a wider variety of fields each year. Recently, neural networks have been tied to a mathematical structure known as *image algebra* [14]. Image algebra was developed specifically for the concise expression and clear representation of image processing techniques and to provide a mathematical environment for image processing algorithm development, comparison, and optimization [5,12,13]. It has been shown that a subalgebra of the image algebra includes the mathematical formulations of currently popular neural network models [14], and image algebra expressions have been derived that represent some well-known algorithms designed for neural network computations. The neural network algorithms represented by these expressions look like their textbook formulations. These image algebra expressions are extremely simple and translucent, and the number of expressions representing each algorithm is very small. Furthermore, the image algebra has suggested a more general concept of neural computation than those that are currently used. In this paper we present a theory for a neural network that uses morphological operations. Several specific applications are given. In particular, we discuss networks that compute the morphological operations of opening and closing. We also give an example of a net that can perform the morphological operation of a sieve. A sieve is a morphological filter which filters out objects which are larger than a specified size [16].

We remark that an entirely different approach to a morphological network was presented in [17]. However, this particular model uses the usual operation of multiplication and summation at each node, a fundamental difference from the model presented here, which uses the operation of addition and maximum at each node.

2. A Brief Overview of the Image Algebra. One appealing aspect of image algebra is its utility as a high level algebraic language for describing image processing algorithms. As has been shown, this language can specify various neural network models in a machine independent way. The operands and operations that define the image algebra have been implemented in the form of a FORTRAN preprocessor, called the *IAF Preprocessor* [1]. Thus, the neural network models found in [14] can be simulated on any machine where the image algebra preprocessor is available. The intent of this paper is to use the concept of a *generalized neural network* introduced in [14] to describe a neural network model which has a distinctly different character than the current linear-based neural networks. The next three sections provide the pertinent image algebra concepts that are necessary to describe the neural network model presented in this paper. A more complete discussion of the image algebra can be found in [5].

2.1. Value Sets and Coordinate Sets. The four basic types of image algebra operands we are interested in are *value sets*, *coordinate sets*, *images*, and *templates*. The value sets considered in this paper are the set of real numbers and the set of extended real numbers, which include one or both of the symbols $-\infty$ and $+\infty$. These are denoted by $\mathbf{R}$, $\mathbf{R}_{-\infty}$, $\mathbf{R}_{+\infty}$, and $\mathbf{R}_{\pm\infty}$, respectively. This corresponds to the values commonly encountered in the type of image processing routines that this paper considers.

An unspecified value set will henceforth be denoted by $\mathbf{F}$. The operations on and between elements of a given value set $\mathbf{F} \in \{\mathbf{R}, \mathbf{R}_{-\infty}, \mathbf{R}_{+\infty}, \mathbf{R}_{\pm\infty}\}$ are the usual elementary operations associated with $\mathbf{F}$. Thus, if $\mathbf{F} = \mathbf{R}$, then the operations are the usual arithmetic and logic operations of addition, multiplication and maximum, and the complementary operations of subtraction, division, and minimum. In addition to the group operation of addition and the semi-group operation of $\vee$ on $\mathbf{R}_{\pm\infty}$, dual operations of $+$ ' and $\wedge$ have been defined on $\mathbf{R}_{\pm\infty}$ in a consistent way such that the resulting structure $\{\mathbf{R}_{\pm\infty}, +, \vee, +', \wedge\}$ mimics many properties of a ring [3]. This has been a critical consideration in the development of the lattice structure of the image algebra. Since mathematical morphology is a subalgebra of the image algebra [6,7,11], a morphological transformation can be viewed as a *lattice transformation* [7]. An arbitrary lattice transform has a representation as a matrix, in context of a special case of a generalized matrix product. The mathematical structure embodying the study of lattice transforms is called *minimax algebra*, which has been primarily used for solving certain types of problems in operations research [3]. The image algebra was used to link morphological image processing operations and more general morphological-type operations to the minimax algebra [7,8]. In addition to these elementary operations on elements of value sets, the image algebra also includes the set operations of union, intersection, (set) subtraction, choice function, and cardinality function on subsets of $\mathbf{F}$. The choice function, when applied to a set, returns a randomly chosen element of that set. Applying the cardinality to a finite set results in an integer which corresponds to the number of elements contained in the set.

Coordinate sets are subsets of n -dimensional Euclidean space $\mathbf{R}^n$. We reserve the letters $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{W}$ to denote coordinate sets.

2.2. Images and Operations on Images. Value sets and coordinate sets constitute some important operands of the image algebra. However, the most fundamental of the algebra's operands are images. Given coordinate and value sets $\mathbf{X}$ and $\mathbf{F}$, respectively, then an $\mathbf{F}$ valued image $\mathbf{a}$ on $\mathbf{X}$ is the graph of a function $\mathbf{a}:\mathbf{X} \rightarrow \mathbf{F}$. Thus, an $\mathbf{F}$ valued image $\mathbf{a}$ on $\mathbf{X}$ is of the form:

$$\mathbf{a} = \{ (\mathbf{x}, \mathbf{a}(\mathbf{x})) : \mathbf{x} \in \mathbf{X}, \mathbf{a}(\mathbf{x}) \in \mathbf{F} \}.$$

The set of all $\mathbf{F}$ valued images on $\mathbf{X}$ is denoted by $\mathbf{F}^{\mathbf{X}}$.

Operations on and between $\mathbf{F}$ valued images are the natural induced operations of the algebraic system $\mathbf{F}$. For example, for real valued images (i.e. elements of $\mathbf{R}^X$), the operations are the elementary operations induced by the vector lattice (a vector space which is also a lattice) $\mathbf{R}$. Thus, the basic real valued image operations reflect the arithmetic and logic operations on $\mathbf{R}$. In particular, the binary operations of image addition, multiplication, and maximum on $\mathbf{R}^X$ are defined as follows:

Let $\mathbf{a}, \mathbf{b} \in \mathbf{R}^X$. Then

$$\mathbf{a} + \mathbf{b} \equiv \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = \mathbf{a}(\mathbf{x}) + \mathbf{b}(\mathbf{x}), \mathbf{x} \in X \} \quad (1)$$

$$\mathbf{a} * \mathbf{b} \equiv \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = \mathbf{a}(\mathbf{x}) * \mathbf{b}(\mathbf{x}), \mathbf{x} \in X \} \quad (2)$$

$$\mathbf{a} \vee \mathbf{b} \equiv \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = \mathbf{a}(\mathbf{x}) \vee \mathbf{b}(\mathbf{x}), \mathbf{x} \in X \} \quad (3)$$

These are the basic binary operations for real valued images. Note that when $\mathbf{F} = \mathbf{R}_{\pm\infty}$, only the operations of addition and maximum can be consistently defined on $\mathbf{R}_{\pm\infty}^X$.

The remaining operations on $\mathbf{R}^X$ or $\mathbf{R}_{\pm\infty}^X$ are induced by the corresponding operations on $\mathbf{R}$ or $\mathbf{R}_{\pm\infty}$, respectively. For example, the *minimum* is defined in terms of the maximum and negation:

$$\mathbf{a} \wedge \mathbf{b} \equiv -(-\mathbf{a} \vee -\mathbf{b}). \quad (4)$$

We also remark that any function $f: \mathbf{F} \rightarrow \mathbf{F}$ induces a function $\mathbf{F}^X \rightarrow \mathbf{F}^X$, again denoted by f , and defined by

$$f(\mathbf{a}) = \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = f(\mathbf{a}(\mathbf{x})) \}$$

For example, $\sin(\mathbf{a}) = \{ (\mathbf{x}, \sin(\mathbf{a}(\mathbf{x}))) : \mathbf{x} \in X \}$. Similarly, if χ_S denotes the generalized characteristic function with respect to some set function $S: X \rightarrow 2^F$, where 2^F denotes the power set of $\mathbf{F}$, then

$$\chi_S(\mathbf{a}) = \{ (\mathbf{x}, \chi_S(\mathbf{a}(\mathbf{x}))) : \mathbf{x} \in X \} = \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = 1 \text{ if } \mathbf{a}(\mathbf{x}) \in S(\mathbf{x}), \text{ otherwise } \mathbf{c}(\mathbf{x}) = 0 \}.$$

We also employ the following notation for some special cases of the generalized characteristic function: Given $\mathbf{a}, \mathbf{b} \in \mathbf{R}^X$, then

$$\chi_{>\mathbf{b}}(\mathbf{a}) \equiv \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = 1 \text{ if } \mathbf{a}(\mathbf{x}) > \mathbf{b}(\mathbf{x}), \text{ else } \mathbf{c}(\mathbf{x}) = 0 \},$$

$$\chi_{\leq \mathbf{b}}(\mathbf{a}) \equiv \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = 1 \text{ if } \mathbf{a}(\mathbf{x}) \leq \mathbf{b}(\mathbf{x}), \text{ else } \mathbf{c}(\mathbf{x}) = 0 \},$$

$$\chi_{\mathbf{b}}(\mathbf{a}) \equiv \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = 1 \text{ if } \mathbf{a}(\mathbf{x}) = \mathbf{b}(\mathbf{x}), \text{ else } \mathbf{c}(\mathbf{x}) = 0 \}, \text{ etc.}$$

If the value set is $\mathbf{R}_{\pm\infty}$, then we may define

$$\chi_{>\mathbf{b}}(\mathbf{a}) \equiv \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = 0 \text{ if } \mathbf{a}(\mathbf{x}) > \mathbf{b}(\mathbf{x}), \text{ else } \mathbf{c}(\mathbf{x}) = -\infty \},$$

$$\chi_{\leq \mathbf{b}}(\mathbf{a}) \equiv \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = 0 \text{ if } \mathbf{a}(\mathbf{x}) \leq \mathbf{b}(\mathbf{x}), \text{ else } \mathbf{c}(\mathbf{x}) = -\infty \},$$

$$\chi_{\mathbf{b}}(\mathbf{a}) \equiv \{ (\mathbf{x}, \mathbf{c}(\mathbf{x})) : \mathbf{c}(\mathbf{x}) = 0 \text{ if } \mathbf{a}(\mathbf{x}) = \mathbf{b}(\mathbf{x}), \text{ else } \mathbf{c}(\mathbf{x}) = -\infty \}, \text{ etc.}$$

As shown in [5], characteristic functions can be defined in terms of the more elementary operations defined earlier.

2.3. Template Operations. In terms of image processing applications, templates and template operations are the most powerful tool of the image algebra. The image algebra definition of a tem-

plate *unifies* and *generalizes* the usual concepts of templates, masks, windows, structuring elements (mathematical morphology), and neighborhood functions into one general mathematical entity [5,7,11].

Let $\mathbf{X}$ and $\mathbf{Y}$ be coordinate sets and $\mathbf{F}$ a value set. A *generalized $\mathbf{F}$ valued template* $\mathbf{t}$ from $\mathbf{Y}$ to $\mathbf{X}$ is a function $\mathbf{t}: \mathbf{Y} \rightarrow \mathbf{F}^{\mathbf{X}}$. Thus $\mathbf{t} \in (\mathbf{F}^{\mathbf{X}})^{\mathbf{Y}}$ and for each $\mathbf{y} \in \mathbf{Y}$, $\mathbf{t}(\mathbf{y}) \in \mathbf{F}^{\mathbf{X}}$; equivalently, for each $\mathbf{y} \in \mathbf{Y}$, $\mathbf{t}(\mathbf{y})$ is an $\mathbf{F}$ valued image on $\mathbf{X}$. The set $\mathbf{Y}$ is called the *target domain* or simply the *domain* of $\mathbf{t}$, and the set $\mathbf{X}$ is called the *range space* of $\mathbf{t}$. For notational convenience we define $\mathbf{t}_{\mathbf{y}} \equiv \mathbf{t}(\mathbf{y})$. Thus, $\mathbf{t}_{\mathbf{y}} = \{(\mathbf{x}, \mathbf{t}_{\mathbf{y}}(\mathbf{x})) : \mathbf{x} \in \mathbf{X}\}$.

If $\mathbf{t} \in (\mathbf{F}^{\mathbf{X}})^{\mathbf{X}}$, then $\mathbf{t}$ is called *translation invariant* if and only if for each triple $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbf{X}$ with $\mathbf{y} + \mathbf{z}$ and $\mathbf{x} + \mathbf{z} \in \mathbf{X}$, we have that $\mathbf{t}_{\mathbf{y}}(\mathbf{x}) = \mathbf{t}_{\mathbf{y}+\mathbf{z}}(\mathbf{x}+\mathbf{z})$. A template which is not translation invariant is called a *translation variant* or, simply, a *variant* template. For example, all morphological structuring elements are translation invariant templates.

If $\mathbf{t} \in (\mathbf{F}^{\mathbf{X}})^{\mathbf{Y}}$, then the *transpose* of $\mathbf{t}$, denoted by $\mathbf{t}'$, is an element of $(\mathbf{F}^{\mathbf{Y}})^{\mathbf{X}}$ and is defined by $\mathbf{t}'_{\mathbf{x}}(\mathbf{y}) = \mathbf{t}_{\mathbf{y}}(\mathbf{x})$. Also, if $\mathbf{t} \in (\mathbf{R}_{\pm\infty}^{\mathbf{X}})^{\mathbf{Y}}$, then the *additive dual* of $\mathbf{t}$ is the template $\mathbf{t}^* \in (\mathbf{R}_{\pm\infty}^{\mathbf{Y}})^{\mathbf{X}}$ defined by

$$\mathbf{t}^*_{\mathbf{x}}(\mathbf{y}) = \begin{cases} -\mathbf{t}'_{\mathbf{x}}(\mathbf{y}) & \text{if } \mathbf{t}_{\mathbf{y}}(\mathbf{x}) \in \mathbf{R} \\ +\infty & \text{if } \mathbf{t}_{\mathbf{y}}(\mathbf{x}) = -\infty \\ -\infty & \text{if } \mathbf{t}_{\mathbf{y}}(\mathbf{x}) = +\infty \end{cases}.$$

All operations between images and templates can be defined via a generalized product between an image and a template. Suppose that $\mathbf{X} \subset \mathbf{R}^n$ is finite, say $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$. If γ is a commutative and associative binary operation on the value set $\mathbf{F}$, then the *global reduce operation* Γ on $\mathbf{F}^{\mathbf{X}}$ induced by γ is defined by

$$\Gamma(\mathbf{a}) = \bigwedge_{\mathbf{x} \in \mathbf{X}} \mathbf{a}(\mathbf{x}) = \mathbf{a}(\mathbf{x}_1) \gamma \mathbf{a}(\mathbf{x}_2) \gamma \dots \gamma \mathbf{a}(\mathbf{x}_m),$$

where $\mathbf{a} \in \mathbf{F}^{\mathbf{X}}$. Thus, $\Gamma: \mathbf{F}^{\mathbf{X}} \rightarrow \mathbf{F}$. In particular, if $\mathbf{F} = \mathbf{R}_{\pm\infty}$ and $\gamma = \vee$, then $\Gamma = \vee$ and $\vee \mathbf{a} = \bigvee_{\mathbf{x} \in \mathbf{X}} \mathbf{a}(\mathbf{x})$.

We combine images and templates by combining appropriate binary operations. Let $\mathbf{F}_1$ and $\mathbf{F}_2$ be two value sets and $o: \mathbf{F}_1 \times \mathbf{F}_2 \rightarrow \mathbf{F}$ a binary operation. If γ is a commutative and associative binary operation on $\mathbf{F}$, $\mathbf{a} \in \mathbf{F}_1^{\mathbf{X}}$, and $\mathbf{t} \in (\mathbf{F}_2^{\mathbf{X}})^{\mathbf{Y}}$, then the *generalized backward template operation* or *generalized right product of $\mathbf{a}$ with $\mathbf{t}$* (induced by γ and o) is the binary operation $\odot: \mathbf{F}_1^{\mathbf{X}} \times (\mathbf{F}_2^{\mathbf{X}})^{\mathbf{Y}} \rightarrow \mathbf{F}^{\mathbf{Y}}$ defined by

$$\mathbf{a} \odot \mathbf{t} \equiv \{(\mathbf{y}, \mathbf{b}(\mathbf{y})) : \mathbf{b}(\mathbf{y}) = \bigwedge_{\mathbf{x} \in \mathbf{X}} \mathbf{a}(\mathbf{x}) o \mathbf{t}_{\mathbf{y}}(\mathbf{x}), \mathbf{y} \in \mathbf{Y}\}.$$

If $\mathbf{t} \in (\mathbf{F}_2^{\mathbf{Y}})^{\mathbf{X}}$ and $o: \mathbf{F}_2 \times \mathbf{F}_1 \rightarrow \mathbf{F}$, then the *generalized forward template operation* or *generalized left product of $\mathbf{a}$ with $\mathbf{t}$* is defined as

$$\mathbf{t} \odot \mathbf{a} \equiv \{(\mathbf{y}, \mathbf{b}(\mathbf{y})) : \mathbf{b}(\mathbf{y}) = \bigwedge_{\mathbf{x} \in \mathbf{X}} \mathbf{t}_{\mathbf{x}}(\mathbf{y}) o \mathbf{a}(\mathbf{x}), \mathbf{y} \in \mathbf{Y}\}.$$

Note that for either the left or right template product, the input image $\mathbf{a}$ is an $\mathbf{F}_1$ valued image on the coordinate set $\mathbf{X}$ while the output image $\mathbf{b}$ is an $\mathbf{F}$ valued image on the coordinate set $\mathbf{Y}$. Thus template operations are capable of transforming images with certain range values defined over a given coordinate set into images with entirely different values defined over coordinate sets of different dimensionality.

There are three basic template operations that are commonly used to transform a real valued image. They are denoted $\oplus$, $\boxplus$, and $\odot$, and called *generalized convolution*, *additive maximum*, and

multiplicative maximum, respectively.

The template operations compute a pixel value $\mathbf{c}(\mathbf{y})$ by performing the basic operation of addition or maximum on a weighted collection of pixel values $\mathbf{a}(\mathbf{x})$, with coordinates $\mathbf{x}$ in some subset of $\mathbf{X}$. Let $\mathbf{X} \subset \mathbf{R}^n$ and $\mathbf{Y} \subset \mathbf{R}^m$ be finite coordinate sets. Let $\mathbf{a} \in \mathbf{R}^{\mathbf{X}}$ and $\mathbf{t} \in (\mathbf{R}^{\mathbf{X}})^{\mathbf{Y}}$. Then $\mathbf{a} \oplus \mathbf{t}$ is defined as

$$\mathbf{a} \oplus \mathbf{t} \equiv \{(\mathbf{y}, \mathbf{c}(\mathbf{y})) : \mathbf{c}(\mathbf{y}) = \sum_{\mathbf{x} \in \mathbf{X}} \mathbf{a}(\mathbf{x}) \cdot \mathbf{t}_{\mathbf{y}}(\mathbf{x}), \mathbf{y} \in \mathbf{Y}\}.$$

For $\mathbf{t} \in (\mathbf{R}_{-\infty}^{\mathbf{X}})^{\mathbf{Y}}$ we define

$$\mathbf{a} \boxplus \mathbf{t} \equiv \{(\mathbf{y}, \mathbf{c}(\mathbf{y})) : \mathbf{c}(\mathbf{y}) = \bigvee_{\mathbf{x} \in \mathbf{X}} \mathbf{a}(\mathbf{x}) + \mathbf{t}_{\mathbf{y}}(\mathbf{x}), \mathbf{y} \in \mathbf{Y}\},$$

and for $\mathbf{t} \in [(\mathbf{R}_{-\infty}^+)^{\mathbf{X}}]^{\mathbf{Y}}$

$$\mathbf{a} \odot \mathbf{t} \equiv \{(\mathbf{y}, \mathbf{c}(\mathbf{y})) : \mathbf{c}(\mathbf{y}) = \bigvee_{\mathbf{x} \in \mathbf{X}} \mathbf{a}(\mathbf{x}) \cdot \mathbf{t}_{\mathbf{y}}(\mathbf{x}), \mathbf{y} \in \mathbf{Y}\}.$$

The additive minimum is defined in terms of duality by

$$\mathbf{a} \boxminus \mathbf{t}^* = -(-\mathbf{a} \boxplus -\mathbf{t}).$$

2.4. Image Algebra and Mathematical Morphology. As mentioned above, mathematical morphology is a subalgebra of the full image algebra. Since our networks will be using the operations of addition and maximum (versus multiplication and summation), we present a basic outline of the relationship between image algebra, mathematical morphology and minimax algebra. The remaining part of this section is discussed in detail in [7,8]

If B represents a structuring element, then a unique translation invariant template $\mathbf{t} \in (\mathbf{R}_{\pm\infty}^{\mathbf{X}})^{\mathbf{X}}$ can be found such that $\mathbf{a} \boxplus \mathbf{t}$ represents the image $\mathbf{a}$ dilated by the structuring element B , and such that $\mathbf{a} \boxminus \mathbf{t}^*$ represents the image $\mathbf{a}$ eroded by the structuring element B . The correspondence is straightforward and is an isomorphism.

The minimax algebra is a matrix theory based on a special case of what is known as a *generalized matrix product* [2]. The latter is defined as follows. Let $\mathbf{F}$ be a set of numbers, f and g be functions from $\mathbf{F} \times \mathbf{F}$ into $\mathbf{F}$, and $\mathbf{F}^{mp}$ be the set of all $m \times p$ matrices with values in $\mathbf{F}$. Let $(a_{ij}) = A \in \mathbf{F}^{mp}$, $(b_{jk}) = B \in \mathbf{F}^{pn}$, and define $f \cdot g$ to be the function from $\mathbf{F}^{mp} \times \mathbf{F}^{pn}$ into $\mathbf{F}^{mn}$ given by

$$(f \cdot g)(A, B) = C, \text{ where} \tag{I}$$

$$c_{ik} = (a_{i1} g b_{1k}) f (a_{i2} g b_{2k}) f \cdots f (a_{ip} g b_{pk}),$$

and f and g are viewed as binary operations. Replacing $\mathbf{F}$ by $\mathbf{R}$ and substituting addition (+) for f and multiplication (*) for g in Equation I, yields the usual matrix product, which corresponds to the image algebra product $\oplus$. Similarly, replacing $\mathbf{F}$ by $\mathbf{R}_{-\infty}$ and substituting the maximum ($\vee$) for f and addition (+) for g results in the minimax matrix product defined in [3]. Thus, a minimax matrix-matrix product is of form

$$c_{ij} = \bigvee_{p=1}^k (a_{ip} + b_{pk})$$

A matrix used in this context is called a *lattice transformation*. The correspondence to the minimax algebra takes a template to a matrix, and an image to a vector. This correspondence is also straightforward as well as an isomorphism. It can be shown that a structuring element is represented by a matrix which is block toeplitz with toeplitz blocks [6], and hence morphological transformations are very special types of lattice transforms.

Due to limited space, we have only presented the most important operands and operations of the image algebra. A detailed description of the full image algebra can be found in [5], and a detailed discussion of the operations of $\boxtimes$ and $\boxdot$ and the relationships between image algebra, minimax algebra, and morphological algebra can be found in [7].

3. Image Algebra and Neural Nets. The relationship between artificial neural networks and the image algebra is discussed in detail in [14]. In this section we give a brief summary of the relevant facts we need, and then present the morphological neural network.

Artificial neural network models are specified by the network topology, node characteristics, and training or learning rules. The two basic equations underlying the theory of computation in a neural network are:

$$\tau_i(t+1) = \sum_{j=1}^n a_j(t) w_{ij} \quad (\text{II})$$

and

$$a_i(t+1) = f(\tau_i(t+1) - \theta), \quad (\text{III})$$

where $a_j(t)$ denotes the value of the j th neuron at time t , n the number of neurons in the network, w_{ij} the synaptic connectivity value between the i th and j th neuron (at time t), $\tau_i(t+1)$ the next or total input effect on the i th neuron, θ a threshold, and f the next state function which usually introduces a nonlinearity into the network. Although not all current network models can be precisely described by these two equations, they nevertheless can be viewed as variations of these.

Letting $\mathbf{X}$ denote the one-dimensional array $\mathbf{X} = \{1, 2, \dots, n\} \subset \mathbf{R}^1$, then $\mathbf{a} \in \mathbf{R}^{\mathbf{X}}$ denotes the set of current values of the neurons (i.e. $\mathbf{a} = \{a(1), a(2), \dots, a(n)\}$, where $a(i)$ denotes the value of the i th neuron), and $\mathbf{b} \in \mathbf{R}^{\mathbf{X}}$ denotes the set of next values. Define $\mathbf{t} \in (\mathbf{R}^{\mathbf{X}})^{\mathbf{X}}$ by $t_i(j) = w_{ij}$ for all $i, j \in \mathbf{X}$. Then Equations II and III correspond to the following equivalent image algebra expressions:

$$\boldsymbol{\tau} = \mathbf{a} \oplus \mathbf{t} \quad (\text{IV})$$

and

$$\mathbf{b} = f(\boldsymbol{\tau} - \theta), \quad (\text{V})$$

where $\boldsymbol{\tau} \in \mathbf{R}^{\mathbf{X}}$ denotes the intermediate image. As a short example, the Hopfield network algorithm in image algebra notation is:

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repeat
     $\mathbf{a} := \mathbf{c}$ 
     $\mathbf{b} := \mathbf{a} \oplus \mathbf{t}$ 
     $\mathbf{c} := 2 * \chi_{>0}(\mathbf{b}) - 1$ 
until  $\mathbf{c} = \mathbf{a}$ 

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Here, $\mathbf{X} = \{1, 2, \dots, n\} \subset \mathbf{R}$, $\mathbf{t} \in (\mathbf{R}^{\mathbf{X}})^{\mathbf{X}}$ represents the weights of the net,

$$t_i(j) = \begin{cases} \sum_{k=1}^m x_j^k x_i^k & i \neq j \\ 0 & i = j \end{cases}$$

x_i^k is the i^{th} element of the exemplar for the pattern class k , and $\mathbf{c} \in \{-1, 1\}^X$ represents the unknown input pattern.

As discussed in [14], image algebra as a mathematical structure encompasses far more general relationships of neural connections and computing than those provided by Equations II and III. In particular, the general form of operations between images (or sets of neurons) and templates (synaptic connection matrices) are provided by the generalized matrix product defined by Equation I. Replacing $\mathbf{F}$ by $\mathbf{R}$ and substituting addition (+) for f and multiplication (*) for g in Equation I, yields the usual matrix product, which corresponds to the image algebra product $\oplus$. This provides for the neural model inherent in Equations II and III or, equivalently, in Equations IV and XIV. Similarly, replacing $\mathbf{F}$ by $\mathbf{R}_{-\infty}$ and substituting the maximum ($\vee$) for f and addition (+) for g results in the minimax matrix product which corresponds to the image algebra product $\boxtimes$. This provides for the novel computational model

$$\tau = \mathbf{a} \boxtimes \mathbf{t} \quad (\text{VI})$$

and

$$\mathbf{b} = f(\tau - \theta) \quad (\text{VII})$$

Of course, Equation VI is also only a special case of Equation I. Substituting the maximum ($\vee$) for f and multiplication (*) for g in Equation V provides for the model

$$\tau = \mathbf{a} \odot \mathbf{t} \quad (\text{VIII})$$

and

$$\mathbf{b} = f(\tau - \theta) \quad (\text{IX})$$

Thus it is clear that Equation I expresses a more general relationship and methods for computing with neural networks than those currently used.

While in classical network models, the initial computation of τ (Equation II) is a linear process, the computation of τ in the model described by Equations VI and VIII is nonlinear. If the neural network model can be expressed using Equations VI and VII as the underlying equations of computation, we call it a *morphological neural network*. The remainder of this paper presents several specific morphological neural networks that can calculate openings, closings, and the boolean sieving filters as described by Serra [16].

The following is a generalized neural network which performs the morphological opening of a gray-valued image $\mathbf{a} \in \mathbf{R}^X$ with the template $\mathbf{t} \in (\mathbf{R}^X)^X$, where $X = \{1, 2, \dots, N\}$. In image algebra notation the opening is denoted by

$$\mathbf{b} = (\mathbf{a} \boxtimes \mathbf{t}^*) \boxtimes \mathbf{t},$$

where $\mathbf{t}$ is the template corresponding to the structuring element g , and $\mathbf{b}$ represents the opened image.

The basic connection scheme of the network is the 2-layer perceptron model [10]. Between the input and output layers, there is one hidden layer of nodes. The weight between input node i and hidden layer node j is denoted by w^{0j} , and the weight between hidden layer node i and output node j is denoted by w^{1j} . The weights are defined by

$$w^{0j} = \begin{cases} -t'_j(i) & \text{if } t'_j(i) \neq -\infty \\ -\infty & \text{otherwise} \end{cases}$$

and

$$w^{1j} = t_j(i) .$$

Define $f^i: \mathbf{R} \rightarrow \mathbf{R}$ to be the activation function (thresholding function) applied at the i -th layer. Here, $f^1: \mathbf{R} \rightarrow \mathbf{R}$ is defined by $f^1(x) = -x$, and $f^2: \mathbf{R} \rightarrow \mathbf{R}$ is the identity function. We will denote the input node values by $A(i)$, the hidden layer values by $H(i)$, and the output layer values by $B(i)$, $i=1, \dots, N$. Initialize the i -th input node value, $A(i)$, to $-a(i)$, $i=1, \dots, N$. In the hidden layer, at node j the node value $H(j)$ is given by

$$H(j) = f^1\left(\sum_{i=1}^N w^{0j} + A(i)\right).$$

At the output layer, the j -th node has value

$$B(j) = f^2\left(\sum_{i=1}^N w^{1j} + H(i)\right) .$$

Then B represents the opening of a with structuring element g .

There are several important items to note about this morphological neural network. First, the only basic morphological operation performed is the dilation. This is, of course, due to the fact that an erosion can be expressed in terms of a dilation. The hidden layer holds the dilation of input image a with the structuring element $-g'$, and the output node values hold the hidden layer values dilated with g . It is well known [16] that *all* morphological operations are combinations of dilations. For example, the fundamental operations of opening and closing are all various combinations of dilations. Second, the weights do not change once they are initialized. That is, there is *no learning* involved. Third, the threshold value θ in Equation VII is 0, that is, there is no threshold value when computing the opening. Fourth, the net stabilizes to the exact desired output after the first pass through the net. Also note that at each iteration, the values of the nodes at the hidden layer and output layer will be finite, even though the calculation at each node involves values of $-\infty$ from the lower weights. This is due to the fact that the input image was finite valued, and that for every $y \in X$, there is at least one $x \in X$ such that $t_y(x)$ is finite valued. Also note that once the weights are initialized, that *any* input applied to the input nodes will result in its opened image at the output nodes. No "retraining" of the weights is necessary, as is sometimes the case in the linear net models.

These are the most important distinctions between the linear-based neural network model and this morphological model.

We can perform a closing in a similar way. In image algebra notation the closing is denoted by

$$b = (a \boxplus t) \boxminus t^*,$$

where t is the template corresponding to the structuring element g , and b represents the closed image.

We use the same network above, with weights defined by

$$w^{0j} = t_j(i),$$

and

$$w^{1j} = \begin{cases} -t'_j(i) & \text{if } t'_j(i) \neq -\infty \\ -\infty & \text{otherwise} \end{cases}.$$

Define $f^i: \mathbf{R} \rightarrow \mathbf{R}$ by $f^i(x) = -x$, $i=1,2$. Initiate the i -th input node value, $A(i)$, to $\mathbf{a}(i)$, $i=1,\dots,N$. In the hidden layer, at node j the node value $H(j)$ is given by

$$H(j) = f^1\left(\sum_{i=1}^N w^{0j} + A(i)\right).$$

At the output layer, the j -th node has value

$$B(j) = f^2\left(\sum_{i=1}^N w^{1j} + H(i)\right).$$

Then B represents the closing of $\mathbf{a}$ with structuring element g .

As in the first net, the only basic morphological operation performed is the dilation. The hidden layer holds the negative of the dilation of input image $\mathbf{a}$ with the structuring element g' , and the output node values hold the negative of the hidden layer values dilated with $-g'$. Again, the weights do not change once they are initialized, and the threshold value is 0. This net too stabilizes to the exact desired output after the first pass through the net.

We call the morphological neural net that performs the opening the *opening net*, and the morphological neural net that performs the closing the *closing net*.

With these two basic operations modeled by a morphological neural network, we can describe other morphological algorithms that use openings or closings by introducing a modified morphological neural network. For example, we can construct a network with performs a sieve filter. A sieve filter has the following form. Let $\mathbf{a}$ be a boolean image on $\mathbf{X}$, and $\mathbf{t}$ a boolean template from $\mathbf{X}$ to $\mathbf{X}$. Here the boolean template $\mathbf{t}$ represents the a shape slightly larger than the object that is desired to be found in the image. Typically, $\mathbf{t}$ is 8-connected and has no "holes" in it, that is, the Euler number of the 8-connected image $\mathbf{t}_y$ is 1. An often-used template is the disk of radius k . Then the image $\mathbf{b}$ is computed, where

$$\mathbf{b} = \mathbf{a} - (\mathbf{a} \boxminus \mathbf{t}^*) \boxminus \mathbf{t}.$$

The image $(\mathbf{a} \boxminus \mathbf{t}^*) \boxminus \mathbf{t}$, the opening of $\mathbf{a}$, will have eroded away the objects of interest, so that when the difference $\mathbf{a} - (\mathbf{a} \boxminus \mathbf{t}^*) \boxminus \mathbf{t}$ is computed, only the objects of interest remain, plus perhaps some noise, which is "smaller" than the size of the desired object. These smaller objects can be eliminated by a similar sieve. We now present a morphological neural net that performs this algorithm.

We use a three-layer neural model to implement the sieve algorithm. We take the opening net, and change the activation function f^2 in the top part from the identity function to the same as f^1 , that is, $f^2(x) = -x$. Then the weights w^{2j} from the second hidden layer to the third layer, which are the output nodes, have value

$$w^{2j}_i = \begin{cases} a(i) & \text{if } i = j \\ -\infty & \text{otherwise} \end{cases}.$$

Here the weight between second hidden layer node i and output layer node j is denoted by w^{2j}_i . The activation function for the third layer, f^3 , is simply the identity function. If we call the node values at the i -th hidden layer by $H_i(i)$, $i=1,2$, then the second hidden layer will have values $H_2(i)$ being the negative of the opened image $\mathbf{a}$. The output layer at node j has value $B(j)$ determined by

$$\begin{aligned} B(j) &= \bigvee_{i=1}^N w^{3j}_i + H_2(i) \\ &= w^{3j}_j + H_2(j) = \mathbf{a}(j) + H_2(j). \end{aligned}$$

It is not difficult to see that $B(j)$ corresponds to the value at the j -th pixel location in $\mathbf{X}$ of the image $\mathbf{b} = \mathbf{a} - (\mathbf{a} \boxtimes \mathbf{t}^*) \boxtimes \mathbf{t}$, the sieved image.

This net has variable weights in the third layer depending on the values of the input image. However, there is no "learning," that is, there is only one pass through the net to arrive at the desired output.

4. Conclusions. The fundamental notions for a theory of morphological neural networks have been presented. Two basic networks, the opening net and the closing net, mimic the morphological operations of opening and closing, respectively, and can be used to build networks that perform sequences of morphological operations, such as the sieving algorithm given in this paper. While the output of this neural network sieving model is no different from the output of the sieving algorithm implemented via morphology, the strength of this theory lies in the fundamentally different approach to neural network computation. The concepts presented in this paper suggest that investigation of generalizations of this network, such as the definition of a learning procedure (a method allowing the change of synaptic weight values) and a back-propagation type method, for example, will undoubtedly result in a neural network model having applications in other areas than pattern recognition. This is an area of future research.

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